

Vehicle Rental System

ER Diagram Attribute Reference

VEHICLECATEGORY

Groups vehicles into types (e.g. Economy, SUV, Luxury).

Attribute	Key	Description
CategoryID	PK	Unique identifier for each category.
CategoryName		The name of the category, such as "SUV" or "Compact".
DailyRate		The price charged per day for renting any vehicle in this category.

VEHICLE

Represents each physical car in the fleet.

Attribute	Key	Description
VehicleID	PK	Unique identifier for each vehicle.
RegistrationNo		The license plate or registration number.
Make		The manufacturer (e.g. Toyota, Ford).
Model		The specific model (e.g. Corolla, Ranger).
Year		The year the vehicle was manufactured.
Status		Current availability, such as Available, Rented, or In Maintenance.
CategoryID	FK	Links the vehicle to its category in VEHICLECATEGORY, which determines its daily rate.

CUSTOMER

Stores the people who rent vehicles.

Attribute	Key	Description
CustomerID	PK	Unique identifier for each customer.
FirstName		The customer's first name.
LastName		The customer's last name.
Phone		Contact phone number.
Email		Contact email address.
DriverLicenseNo		The customer's driver's license number, used to verify they are legally allowed to drive.

USER

Handles system login accounts (people who log into the application).

Attribute	Key	Description
UserID	PK	Unique identifier for each user account.
Username		The login name.
PasswordHash		The password stored in encrypted (hashed) form, never as plain text, for security.
Role		The permission level, such as Admin, Staff, or Customer.
CustomerID	FK	Links the account to a customer record, so a customer can log in and see their own bookings.

BOOKING

Records each rental reservation. It is the central table connecting customers and vehicles.

Attribute	Key	Description
BookingID	PK	Unique identifier for each booking.
CustomerID	FK	Which customer made the booking.
VehicleID	FK	Which vehicle was booked.
StartDate		The date the rental begins.
EndDate		The date the rental ends or the vehicle is due back.
BookingStatus		The state of the booking, such as Pending, Confirmed, Active, Completed, or Cancelled.

PAYMENT

Tracks money paid for bookings.

Attribute	Key	Description
PaymentID	PK	Unique identifier for each payment.
BookingID	FK	Which booking the payment is for.
Amount		The amount paid.
PaymentDate		When the payment was made.
PaymentMethod		How it was paid, such as Cash, Card, or EFT.

Key Terms

PK (Primary Key): Uniquely identifies each row in a table.

FK (Foreign Key): References a primary key in another table, creating the relationship between them.

Relationships

Relationship	Cardinality	Meaning
VEHICLECATEGORY - VEHICLE	1 : M	One category has many vehicles.

Relationship	Cardinality	Meaning
VEHICLE - BOOKING	1 : M	One vehicle can be booked many times (over different dates).
CUSTOMER - BOOKING	1 : M	One customer can make many bookings.
BOOKING - PAYMENT	1 : M	One booking can have multiple payments (e.g. deposit plus final payment).
USER - CUSTOMER	1 : 1	Each user account belongs to exactly one customer, and vice versa.